

# Design a Unique ID Generator - Step 4: Wrap Up

## 1. Approaches Discussed

- Multi-master replication
- UUID
- Ticket server
- Snowflake-like ID generator by Twitter

## 2. Final Decision

Snowflake approach is selected

- Supports all requirements
- Works well in **distributed systems**
- Highly **scalable**

## 3. Additional Talking Points

### Clock Synchronization

- Assumes all servers have the **same clock**
- Not always true in:
  - Multi-core systems
  - Multi-machine environments

Solution:

- Use **Network Time Protocol (NTP)**

## Section Length Tuning

- Bit allocation can be adjusted based on use case

Example:

- More timestamp bits → longer lifespan
- Fewer sequence bits → lower concurrency

## High Availability

- ID generator is a **mission-critical system**
- Must ensure:
  - High availability
  - Fault tolerance

## 4. Key Takeaway

Snowflake:

- Best fit for:
  - Distributed systems
  - High scale
  - Ordered unique ID generation